

Algebra 1
Unit 8
Lesson 9 Practice

Name Answer Key
Date _____
Period _____

Florida Real Estate Team just bought a 100 unit apartment building and plans to rent all of the apartments out. The function $P(u) = -10u^2 + 1760u - 50,000$ represents the profit earned, P , depending on the amount of apartment units, u , they rent.

1. Rewrite the function in vertex form.

$$P(u) = -10(u^2 - 176 + 7,744) - 50,000 + 77,400$$

$$P(u) = -10(u - 88)^2 + 27,440$$

Complete the statements.

2. The

☐ vertex
☒ factored
☐ standard

 form of the function could be used to determine the interval on which the real estate company would experience a profit.

3. The

☒ vertex
☐ factored
☐ standard

 form of the function can be used to determine when the real estate company experiences the greatest profit.

4. The company is predicted to experience the greatest profit when 88 units are rented, resulting in a profit of \$ 27,440.

Adam is coding a new game where darts are launched at a moving target. He enters the function $H(t) = -0.5t^2 + 6t - 10$ to model the height, H , of the darts in relationship to the ground after t seconds.

5. Rewrite the function in vertex form.

$$H(t) = -0.5(t^2 - 12t + 36)^2 - 10 + 18$$

$$H(t) = -0.5(t - 6)^2 + 8$$

6. Rewrite the function in factored form.

$$H(t) = -0.5(t^2 + 12t + 20)$$

$$H(t) = -0.5(t - 2)(t - 10)$$

Complete the statements.

7. The ☐ vertex
☒ factored
☐ standard form of the function can be used to determine how long the

dart is above the ground. The dart is in the air for 2 < t < 10, 8 seconds

8. The dart was launched from a point 10 units ☐ above
☒ below the ground and

went ☒ above
☐ below ground after 2 seconds.

9. The ☒ vertex
☐ factored
☐ standard form of the function can be used to determine when the

dart reaches its highest point.

10. The dart reaches its highest point of 8 units 6 seconds after it is launched.